Haotian Zeng

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Go, Python, Java, Bash, HTML/CSS/JavaScript, Ruby, C, C++, Groovy
PostgreSQL, gRPC, Terraform, Prometheus, AWS, Azure, Docker, Jenkins

PROJECTS

Job App Tracker with Lootboxes and Agentic Functionality Apr 2025 - current

- Designed and implemented full-scale REST web application with gamified streak and rewards tracking system, leveraging AI tools to expedite the learning, architecture planning, and design iteration process.
- Developing an AI feature that summarizes job and role description from web-scraped job post urls.

SSH Certificate Injection (Hashicorp) Dec 2022 – Jan 2023

- Built the first major Boundary Enterprise feature enabling direct SSH key injection into Vault, resulting in a more secure and efficient user experience.
- Collaborated cross-functionally to split a complex feature into deliverable components, resulting in the successful launch of a major Boundary enterprise feature within the financial quarter.

Boundary Metrics (Hashicorp) Feb 2022 – March 2022

- Implemented server and client-side metrics (latency, request size, connection count, and connection status) for gRPC, HTTP, and WebSocket protocols.
- Authored the initial design document, gathered feedback from senior engineers, and iterated quickly to swiftly gain domain expertise and advance the development process.
- Expanded feature scope by refactoring codebase, adding documentation and tests for new metrics and connection types, as the project evolved.

Monte Carlo Uncertainty Quantifier (Lawrence Berkeley Laboratory) June 2021 – Nov 2021

- Automated PHREEQC simulations to study uranium seepage in groundwater, analyzing outputs with Pandas and Pyplot.
- Built a reproducible computational pipeline enabling large-scale geochemical sensitivity analyses.

Dragonfly Pipeline (Wind River) June 2019 – Dec 2019

- Integrated Jenkins, Python, and Bash to build a CI tool for core product testing, cutting test cycles by hours.
- Automated dynamic test selection menus with cron jobs, improving usability and reliability.

EXPERIENCE

Hashicorp – Backend Engineer (Boundary) Dec 2021 – June 2023

- Designed and shipped the Metrics Package, providing the Cloud Team with insights into thousands of live clusters, facilitating real-time diagnosis and ensuring 99.9% uptime for customer clusters, thereby enhancing revenue streams.
- Developed the Vault SSH Certificate Injection feature, streamlining user authentication workflows and improving session authorization flow by 20%.
- Refactored the PostgreSQL database, programmatically renaming thousands of tables and constraints to improve scalability and consistency.
- Delivered Controller Metadata functionality, providing essential license metadata to a Front-End API endpoint for versioning of Enterprise products.
- Led release candidate testing, coordinated cross-functional testing efforts, and authored comprehensive test cases, ensuring software quality at all stages.

Lawrence Berkeley National Laboratory – Research Assistant June 2021 – Nov 2021

- Designed an intuitive wrapper using MVC principles to launch geochemistry simulation software with complex input parameters, resulting in plottable data outputs.
- Employed sensitivity analysis and machine learning optimization algorithms on diverse datasets, contributing to a peer-reviewed manuscript publication.

Wind River Systems – Software Engineering Intern June 2019 – Dec 2019

- Developed a CI tool to specify complex test cases and parameters, significantly increasing testing pipeline speed by up to 6 hours per run.
- Implemented query tool for Sales team, revitalizing access to a legacy database and enabling new sales opportunities.

EDUCATION

University of California, Berkeley – Computer Science, Nuclear Engineering

Relevant Courses: Data Structures, Computer Architecture, Probability, Computer Security, Operating Systems & Systems Programming, Artificial Intelligence, Linear Algebra, Multivariable Calculus